PROJECT PROPOSAL

SkillLens

A Skill-to-Employment Intelligence Platform for Indian Engineering Students

Project	SkillLens — Skill-to-Employment Intelligence Platform
Institute	Swami Vivekanand College of Engineering, Indore
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Date	March 2026

1. Executive Summary

SkillLens is a web-based career preparation platform designed specifically for Indian engineering students preparing for campus placements. The platform moves beyond keyword-based resume scanning — the current state of most tools — and introduces skill depth measurement: a system that quantifies how well a student knows each skill, maps that depth to real company requirements, and generates a personalised, resource-linked weekly preparation roadmap.

The project is proposed as a Final Year Major Project by three B.E. Computer Science students at Swami Vivekanand College of Engineering, Indore, under the mentorship of Ms. Amisha Jain. The system will be built using the MERN stack (MongoDB, Express.js, React.js, Node.js) with OpenAI API integration for interview evaluation.

2. Problem Statement

Indian engineering graduates face a structured set of problems during placement preparation that existing platforms do not adequately address:

2.1 Skill presence vs. skill depth

Current resume tools detect whether a skill keyword exists in a resume. They cannot determine how well the student actually knows that skill. A student who used React once in a one-week project and a student who has built three production applications both receive the same resume score. This makes the score meaningless for both the student and the employer.

2.2 No company-specific gap analysis

Students apply to companies without knowing which specific skills they are deficient in for that company's requirements. A student might be 78% ready for TCS but only 31% ready for Razorpay — and the gap is entirely in MongoDB depth and DSA. No existing platform provides this level of targeted, company-specific insight.

2.3 Communication skills ignored in technical interviews

A significant proportion of engineering graduates in Tier 2 and Tier 3 cities lose interviews not due to technical gaps but due to poor communication — inability to structure an answer clearly, use correct terminology, or demonstrate confidence. No existing mock interview tool evaluates communication quality separately from technical correctness.

2.4 Isolated tools, no integrated preparation system

Students currently use multiple disconnected tools: one for resume checking, another for mock interviews, another for DSA practice. No single platform connects these activities into a feedback loop where improving in one area automatically updates their readiness assessment across all others.

3. Proposed Solution

SkillLens proposes five tightly integrated modules that form a continuous improvement loop:

Module	What it does	Key innovation
Smart Resume Analyzer	Extracts skills from uploaded PDF resume using text parsing	Triggers a skill depth quiz per detected skill — not just keyword detection
Skill Depth Quiz Engine	10 MCQs per skill, randomised, scored as a depth percentage	First time skill depth (not presence) is measured in a placement prep tool
Company Match	Maps student's depth scores to	Weighted gap analysis — shows exactly

Engine	25 real Indian company profiles	which skill to improve first for a target company
AI Mock Interview	Role-specific questions evaluated by OpenAI	Dual scoring: technical accuracy + communication quality, with structured actionable feedback
Placement Roadmap	Week-by-week plan with curated free resources	AI decides order based on gaps; actual resources come from a curated JSON library — specific, clickable, real

All five modules feed into a single Readiness Score (0–100) that updates every time the student completes any activity. This score is the north star metric — it answers 'am I ready for placement?' in one number, with a breakdown of exactly what is and is not contributing to it.

4. Project Objectives

1. To build a skill depth measurement system that goes beyond keyword detection and quantifies actual knowledge level per skill.
2. To create a company eligibility engine using real Indian company job requirements, giving students specific, company-named targets.
3. To implement an AI-powered mock interview system that evaluates both technical and communication dimensions of an answer.
4. To generate personalised, resource-linked weekly preparation roadmaps driven by each student's specific skill gaps.
5. To synthesise all module outputs into a single Readiness Score that tracks improvement over time.
6. To build a platform that is genuinely free for students, with a monetisation path via college institutional plans.

5. Target Users

User	Profile	Primary Need
Final-year engineering student	6–12 months from placement season, CS or IT branch	Know exactly which skills to improve and for which companies
Pre-final-year student	1–2 years from placement, wants early preparation	Build a structured roadmap from current skill level
Placement cell coordinator	College admin managing placement activities	Understand cohort-level readiness and identify weakest areas across the batch

6. Uniqueness and Market Gap

A review of available platforms reveals the following landscape:

Platform	What it offers	What it lacks
Naukri / LinkedIn	Job listings, resume upload, basic ATS score	No skill depth measurement, no interview practice, no roadmap
Resumod / Zety	Resume formatting and keyword scoring	No company-specific matching, no interview, no learning plan
HackerRank / LeetCode	Coding practice and assessments	No resume analysis, no company matching, no communication feedback

InterviewBit	DSA practice with company tags	No resume analysis, no communication scoring, no roadmap generation
SkillLens (proposed)	All of the above — integrated into one feedback loop	Nothing comparable exists for Indian students in this integrated form

No platform currently combines skill depth measurement, weighted company matching against real Indian companies, dual-dimension interview scoring, and a curated resource-linked roadmap in a single integrated system.

7. Proposed Technology Stack

Layer	Technology	Justification
Frontend	React.js, Tailwind CSS, Recharts	Component-based UI; Recharts for dashboard visualisations
Backend	Node.js, Express.js	Fast, non-blocking; team has existing proficiency
Database	MongoDB + Mongoose	Flexible schema suits varied data structures per module
AI Layer	OpenAI API (gpt-3.5-turbo)	Used only for interview evaluation — controlled, cost-aware usage
Auth	JWT + bcrypt	Industry-standard stateless authentication
File Handling	Multer + pdf-parse	Resume upload and text extraction
Deployment	Vercel (frontend), Render (backend)	Both free tier; sufficient for project demonstration

8. Expected Deliverables

- Fully functional web application with all five modules operational
- Company profile JSON covering 25 Indian companies — manually curated from real job postings
- Skill quiz question bank — 100 questions across 10 skills
- Resource library JSON — curated videos, articles, practice links, and tasks per skill per level
- Complete project documentation set: SRS, System Design, API Design, Test Plan, Final Report
- Deployed application accessible via public URL

9. Mentor Approval

Field	Details
Mentor Name	Ms. Amisha Jain
Department	Computer Science & Engineering
Approval Status	Pending / Approved
Date	March 2026
Signature	